

Decision Tree – Entropy, Information Gain, Gini Index

Instance	a_1	a_2	a_3	Target Class
1	T	T	1.0	+
2	T	T	6.0	+
3	T	F	5.0	-
4	F	F	4.0	+
5	F	T	7.0	-
6	F	T	3.0	-
7	F	F	8.0	-
8	T	F	7.0	+
9	F	T	5.0	-

1. The entropy of the training examples is

$$Entropy(S) = - \sum_{i=1}^n p_i \log_2(p_i)$$

$$Entropy(S) = - \frac{4}{9} \log_2\left(\frac{4}{9}\right) - \frac{5}{9} \log_2\left(\frac{5}{9}\right)$$

+ve -ve

$$= 0.9911$$

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1	T	T	1.0	+
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3	T	F	5.0	-
4	F	F	4.0	+
5	F	T	7.0	-
6	F	T	3.0	-
7	F	F	8.0	-
8	T	F	7.0	+
9	F	T	5.0	-

2. What is the information gain of the a_1 with respect to the training examples.

$$Entropy(S) = - \sum_{i=1}^n p_i \log_2(p_i)$$

$$Entropy(S_T) = - \frac{3}{4} \log_2\left(\frac{3}{4}\right) - \frac{1}{4} \log_2\left(\frac{1}{4}\right)$$

$$= 0.311 + 0.5 = 0.811$$

$$Entropy(S_F) = - \frac{1}{5} \log_2\left(\frac{1}{5}\right) - \frac{4}{5} \log_2\left(\frac{4}{5}\right)$$

$$= 0.4644 + 0.2576 = 0.722$$

$a_1 = T \quad a_1 = F$

+ve	3	2
-ve	1	4

- i) find entropy of pred (tot entropy)
- ii) find entropy of attrb1 (for both T & F)
- iii) info gain of attrb1 (tot entropy - prob of T x entropy of T in a_1 - prob of F x entropy of F in a_1)

iv) gini of attrb1 (T & F)

Decision Tree – Entropy, Information Gain, Gini Index

Instance	a_1	a_2	a_3	Target Class
1	T	T	1.0	+
2	T	T	6.0	+
3	T	F	5.0	-
4	F	F	4.0	+
5	F	T	7.0	-
6	F	T	3.0	-
7	F	F	8.0	-
8	T	F	7.0	+
9	F	T	5.0	-

2. What is the information gain of the a_1 with respect to the training examples.

$$Gain(a_1) = \underbrace{Entropy(S)}_{(+)} - \sum_{v \in \{T, F\}} \underbrace{\frac{|S_v|}{|S|}}_{a_1 \text{ weight}} \underbrace{Entropy(S_v)}_{calc entropy}$$

$$Gain(a_1) = Entropy(S) - \frac{4}{9} Entropy(S_T) - \frac{5}{9} Entropy(S_F)$$

(acc to a_1 only)

$$Gain(a_1) = 0.9911 - \frac{4}{9} * 0.811 - \frac{5}{9} * 0.722 = 0.2295$$

Instance	a_1	a_2	a_3	Target Class
1	T	T	1.0	+
2	T	T	6.0	+
3	T	F	5.0	-
4	F	F	4.0	+
5	F	T	7.0	-
6	F	T	3.0	-
7	F	F	8.0	-
8	T	F	7.0	+
9	F	T	5.0	-

3. Compute the Gini Index of the attributes a_1 .

$$Gini = 1 - \sum_{i=1}^n (p_i)^2$$

$$Gini(T) = 1 - \left(\frac{3}{4}\right)^2 - \left(\frac{1}{4}\right)^2 = 0.375$$

(Toussaint found me)

$$Gini(F) = 1 - \left(\frac{1}{5}\right)^2 - \left(\frac{4}{5}\right)^2 = 0.32$$

Instance	a_1	a_2	a_3	Target Class
1	T	T	1.0	+
2	T	T	6.0	+
3	T	F	5.0	-
4	F	F	4.0	+
5	F	T	7.0	-
6	F	T	3.0	-
7	F	F	8.0	-
8	T	F	7.0	+
9	F	T	5.0	-

3. Compute the Gini Index of the attributes a_1 .

$$GiniIndex(a_1) = \sum_{v \in \{T, F\}} \frac{|S_v|}{|S|} Gini(S_v)$$

weight of each class

$$GiniIndex(a_1) = \left(\frac{4}{9}\right) * Gini(T) + \left(\frac{5}{9}\right) * Gini(F)$$

$$GiniIndex(a_1) = \left(\frac{4}{9}\right) * 0.375 + \left(\frac{5}{9}\right) * 0.32$$

$$GiniIndex(a_1) = \underline{\underline{0.3444}}$$

Decision Tree – Entropy, Information Gain, Gini Index

Instance	a_1	a_2	a_3	Target Class
1	T	T	1.0	+
2	T	T	6.0	+
3	T	F	5.0	-
4	F	F	4.0	+
5	F	T	7.0	-
6	F	T	3.0	-
7	F	F	8.0	-
8	T	F	7.0	+
9	F	T	5.0	-

3. Compute the Gini Index of the attributes a_2 .

$$Gini = 1 - \sum_{i=1}^n (p_i)^2$$

$$Gini(T) = 1 - \left(\frac{2}{5}\right)^2 - \left(\frac{3}{5}\right)^2 = \underline{\underline{0.48}}$$

$$Gini(F) = 1 - \left(\frac{2}{4}\right)^2 - \left(\frac{2}{4}\right)^2 = 0.5$$

Decision Tree – Entropy, Information Gain, Gini Index

Instance	a_1	a_2	a_3	Target Class
1	T	T	1.0	+
2	T	T	6.0	+
3	T	F	5.0	-
4	F	F	4.0	+
5	F	T	7.0	-
6	F	T	3.0	-
7	F	F	8.0	-
8	T	F	7.0	+
9	F	T	5.0	-

3. Compute the Gini Index of the attributes a_2 .

$$GiniIndex(a_2) = \sum_{v \in \{T, F\}} \frac{|S_v|}{|S|} Gini(S_v)$$

$$GiniIndex(a_2) = \left(\frac{5}{9}\right) * \underline{Gini(T)} + \left(\frac{4}{9}\right) * \underline{Gini(F)}$$

$$GiniIndex(a_2) = \left(\frac{5}{9}\right) * 0.48 + \left(\frac{4}{9}\right) * 0.5$$

$$GiniIndex(a_2) = \underline{0.4889}$$

Decision Tree – Entropy, Information Gain, Gini Index

Instance	a_1	a_2	a_3	Target Class
1	T	T	1.0	+
2	T	T	6.0	+
3	T	F	5.0	-
4	F	F	4.0	+
5	F	T	7.0	-
6	F	T	3.0	-
7	F	F	8.0	-
8	T	F	7.0	+
9	F	T	5.0	-

4. Which is the best splitting attribute between a_1 and a_2 .

$$Gain(a_1) = \underline{0.2295}$$

$$Gain(a_2) = \underline{0.0072}$$

Higher Information Gain Produces Better Split

Hence, attribute a_1 is the best split attribute

* Entropy, I.G $\Rightarrow > \Rightarrow$ better split attr

* G. Index $\Rightarrow < \Rightarrow$ better split attr

Decision Tree – Entropy, Information Gain, Gini Index

Instance	a_1	a_2	a_3	Target Class
1	T	T	1.0	+
2	T	T	6.0	+
3	T	F	5.0	-
4	F	F	4.0	+
5	F	T	7.0	-
6	F	T	3.0	-
7	F	F	8.0	-
8	T	F	7.0	+
9	F	T	5.0	-

4. Which is the best splitting attribute between a_1 and a_2 .

$$\text{GiniIndex}(a_1) = \underline{0.3444}$$

$$\text{GiniIndex}(a_2) = \underline{0.4889}$$

Smaller GiniIndex Produces Better Split

Hence, attribute a_1 is the best split attribute

Decision Tree – Entropy, Information Gain, Gini Index

A	B	Class Label
T	F	+
T	T	+
T	T	+
T	F	-
T	T	+
F	F	-
F	F	-
F	F	-
T	T	-
T	F	-

The contingency tables after splitting on attributes A and B are:

	$A = T$	$A = F$		$B = T$	$B = F$
+	4	0	+	3	1
-	3	3	-	1	5

The overall entropy before splitting is:

$$E_{orig} = -0.4 \log_2 0.4 - 0.6 \log_2 0.6 = \underline{0.9710}$$

The information gain after splitting on A is:

$$E_{A=T} = -\frac{4}{7} \log_2 \frac{4}{7} - \frac{3}{7} \log_2 \frac{3}{7} = \underline{0.9852}$$

$$E_{A=F} = -\frac{3}{3} \log_2 \frac{3}{3} - \frac{0}{3} \log_2 \frac{0}{3} = \underline{0}$$

$$\Delta = E_{orig} - \frac{7}{10} E_{A=T} - \frac{3}{10} E_{A=F} = 0.2813$$

$$E(S) = - \sum_{i=1}^n p_i \log_2(p_i)$$

(i) Take for each attribute

(ii) Tot entropy

$$p_+ = \frac{4}{7} = 0.4$$

$$p_- = \frac{3}{7} = 0.6$$

(iii) entropy of 'A'

of 'A'

(iv) info gain of 'A'

tot w (calc) w (calc)

Decision Tree – Entropy, Information Gain, Gini Index

A	B	Class Label
T	F	+
T	T	+
T	T	+
T	F	-
T	T	+
F	F	-
F	F	-
F	F	-
T	T	-
T	F	-

(iv) entropy of B, info gain of B :

The information gain after splitting on B is:

$$E_{B=T} = -\frac{3}{4} \log \frac{3}{4} - \frac{1}{4} \log \frac{1}{4} = 0.8113$$

$$E_{B=F} = -\frac{1}{6} \log \frac{1}{6} - \frac{5}{6} \log \frac{5}{6} = 0.6500$$

$$\Delta = E_{orig} - \frac{4}{10} E_{B=T} - \frac{6}{10} E_{B=F} = 0.2565$$

Therefore, attribute A will be chosen to split the node

Best split attr??
A

Entropy overall: (same for all attr)

$$-p_i \log_2(p_i) \dots$$

for predic

Entropy of attr:

$$E_{attr=val1} = -p_i \log_2(p_i) - p_j \log_2(p_j)$$

$$E_{attr=val2}$$

Info gain:

$$\text{Tot entropy} = \text{weight of val1} \times \text{entropy of val1} + \text{weight of val2} \times \text{entropy of val2}$$

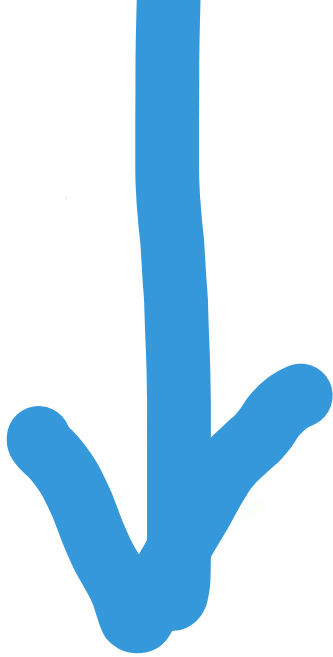
($E_{attr=val1}$) ($E_{attr=val2}$)

Gini:

$$\text{val1} \quad 1 - (p_i)^2 \dots$$

$$\text{val2} \quad 1 - (p_j)^2 \dots$$

$$\text{Gini index} = \text{weight of } v_1 \times \text{gini val1} + \text{weight of } v_2 \times \text{gini val2}$$



ENTROPY AND INFORMATION GAIN

Day	Outlook	Temp	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes

Values(Wind) = Weak, Strong

$$S = [9+, 5-]$$

$$S_{Weak} \leftarrow [6+, 2-]$$

$$S_{Strong} \leftarrow [3+, 3-]$$

$$\begin{aligned} \text{Gain}(S, \text{Wind}) &= \text{Entropy}(S) - \sum_{v \in \{\text{Weak}, \text{Strong}\}} \frac{|S_v|}{|S|} \text{Entropy}(S_v) \\ &= \text{Entropy}(S) - (8/14) \text{Entropy}(S_{Weak}) \\ &\quad - (6/14) \text{Entropy}(S_{Strong}) \\ &= 0.940 - (8/14)0.811 - (6/14)1.00 \\ &= 0.048 \end{aligned}$$

Handwritten table for Wind values:

	Weak	Strong
yes	6	3
no	2	3

Decision Tree using Gini Index:

- * we have to select one attr having max information types.
- * calc total gini index of predic class.

Decision Tree using Gini Index – Solved Example

Weekend	Weather	Parents	Money	Decision
W1	Sunny	Yes	Rich	Cinema
W2	Sunny	No	Rich	Tennis
W3	Windy	Yes	Rich	Cinema
W4	Rainy	Yes	Poor	Cinema
W5	Rainy	No	Rich	Stay In
W6	Rainy	Yes	Poor	Cinema
W7	Windy	No	Poor	Cinema
W8	Windy	No	Rich	Shopping
W9	Windy	Yes	Rich	Cinema
W10	Sunny	No	Rich	Tennis

- Compute the **Gini Index** for the overall collection of training examples.
- There are **four possible output variables** Cinema, Tennis, Stay In and Shopping.
- The data has **6 instances of Cinema**, **2 instances of Tennis**, 1 instance of Stay In and 1 of shopping.

$$Gini(S) = 1 - \left[\left(\frac{6}{10} \right)^2 + \left(\frac{2}{10} \right)^2 + \left(\frac{1}{10} \right)^2 + \left(\frac{1}{10} \right)^2 \right] = 0.58$$

(Handwritten note: 6 Cinema, 2 Tennis, 1 Stay In, 1 Shopping)

- * calc gini index for each attr & construct the tree.

Decision Tree using Gini Index – Solv

Weekend	Weather	Parents	Money	Decision
W1	Sunny	Yes	Rich	Cinema
W2	Sunny	No	Rich	Tennis
W3	Windy	Yes	Rich	Cinema
W4	Rainy	Yes	Poor	Cinema
W5	Rainy	No	Rich	Stay In
W6	Rainy	Yes	Poor	Cinema
W7	Windy	No	Poor	Cinema
W8	Windy	No	Rich	Shopping
W9	Windy	Yes	Rich	Cinema
W10	Sunny	No	Rich	Tennis

- Computation of **Gini Index for Money Attribute**
- It has **two possible values of Rich (7 examples)** and **Poor (3 examples)**.
- For **Money = Poor**, there are **3 examples with "Cinema"**.
- $Gini(S) = 1 - \left[\left(\frac{3}{3} \right)^2 \right] = 0$ ✓
- For **Money = Rich**, there are **2 examples with "Tennis"**, **3 examples with "Cinema"** and **1 example with "Stay in", "Shopping" each**
- $Gini(S) = 1 - \left[\left(\frac{2}{7} \right)^2 + \left(\frac{3}{7} \right)^2 + \left(\frac{1}{7} \right)^2 + \left(\frac{1}{7} \right)^2 \right] = 0.694$
- **Weighted Average (Money)**

$$= 0 \cdot \left(\frac{3}{10} \right) + 0.694 \cdot \left(\frac{7}{10} \right) = 0.486$$

$$Gini_{M=P} = 1 - \left(\frac{3}{3} \right)^2 = 0$$

$$Gini_{M=R} = 1 - \left(\frac{2}{7} \right)^2 - \left(\frac{3}{7} \right)^2 - \left(\frac{1}{7} \right)^2 - \left(\frac{1}{7} \right)^2 = 0.694$$

$$Gini Index = \left(\frac{3}{10} \times 0 \right) + \left(\frac{7}{10} \times 0.694 \right) = 0.486$$

* My for parents attr:

Decision Tree using Gini Index – Solved Example

Weekend	Weather	Parents	Money	Decision
W1	Sunny	Yes	Rich	Cinema
W2	Sunny	No	Rich	Tennis
W3	Windy	Yes	Rich	Cinema
W4	Rainy	Yes	Poor	Cinema
W5	Rainy	No	Rich	Stay In
W6	Rainy	Yes	Poor	Cinema
W7	Windy	No	Poor	Cinema
W8	Windy	No	Rich	Shopping
W9	Windy	Yes	Rich	Cinema
W10	Sunny	No	Rich	Tennis

- Computation of **Gini Index for Parents Attribute**
- It has two possible values of **Yes (5 examples)** and **No (5 examples)**.
- For **Parents = Yes**, there are **5 examples**, all with "Cinema".
- $Gini(S) = 1 - \left[\left(\frac{5}{5} \right)^2 \right] = 0$
- For **Parents = No**, there are **2 examples with "Tennis"**, **1 example with "Stay in", "Shopping" and "Cinema" each**
- $Gini(S) = 1 - \left[\left(\frac{2}{5} \right)^2 + \left(\frac{1}{5} \right)^2 + \left(\frac{1}{5} \right)^2 + \left(\frac{1}{5} \right)^2 \right] = 0.72$
- **Weighted Average (Parents)**

$$= 0 \cdot \left(\frac{5}{10} \right) + 0.72 \cdot \left(\frac{5}{10} \right) = 0.36$$

* My for weather attr:

Decision Tree using Gini Index – Solved Example

Weekend	Weather	Parents	Money	Decision
W1	Sunny	Yes	Rich	Cinema
W2	Sunny	No	Rich	Tennis
W3	Windy	Yes	Rich	Cinema
W4	Rainy	Yes	Poor	Cinema
W5	Rainy	No	Rich	Stay In
W6	Rainy	Yes	Poor	Cinema
W7	Windy	No	Poor	Cinema
W8	Windy	No	Rich	Shopping
W9	Windy	Yes	Rich	Cinema
W10	Sunny	No	Rich	Tennis

- Computation of **Gini Index for Weather Attribute**
- It has three possible values of **Sunny (3 examples)**, **Rainy (3 examples)** and **Windy (4 examples)**.
- For **Weather = Sunny**, there are **2 examples with "Cinema"** and 1 with "Tennis".
- $Gini(Sunny) = 1 - \left[\left(\frac{2}{3} \right)^2 + \left(\frac{1}{3} \right)^2 \right] = 0.444$
- For **Weather = Rainy**, there are **2 examples with "Cinema"** and 1 example with "Stay in"
- $Gini(Rainy) = 1 - \left[\left(\frac{2}{3} \right)^2 + \left(\frac{1}{3} \right)^2 \right] = 0.444$
- For **Weather = Windy**, there are **3 examples with "Cinema"** and 1 example with "Shopping"
- $Gini(Windy) = 1 - \left[\left(\frac{3}{4} \right)^2 + \left(\frac{1}{4} \right)^2 \right] = 0.375$

$$Gini Index = \left(\frac{3}{10} \times 0.444 \right) + \left(\frac{3}{10} \times 0.444 \right) + \left(\frac{4}{10} \times 0.375 \right)$$

* Gini index for weather = 0.416
parents = 0.36
money = 0.486

Smallest Gini index $\Rightarrow$ Parents

Decision Tree using Gini Index – Solve

Parents				
Yes				
Weekend	Weather	Parents	Money	Decision
W1	Sunny	Yes	Rich	Cinema
W3	Windy	Yes	Rich	Cinema
W4	Rainy	Yes	Poor	Cinema
W6	Rainy	Yes	Poor	Cinema
W9	Windy	Yes	Rich	Cinema
No				
Weekend	Weather	Parents	Money	Decision
W2	Sunny	No	Rich	Tennis
W5	Rainy	No	Rich	Stay In
W7	Windy	No	Poor	Cinema
W8	Windy	No	Rich	Shopping
W10	Sunny	No	Rich	Tennis

no need to make changes

divide into (i) yes
(ii) no

again from weather & money we need to Gini index:

Decision Tree using Gini Index – Solved Example

Weekend	Weather	Parents	Money	Decision
W2	Sunny	No	Rich	Tennis
W5	Rainy	No	Rich	Stay In
W7	Windy	No	Poor	Cinema
W8	Windy	No	Rich	Shopping
W10	Sunny	No	Rich	Tennis

Computation of Gini Index for Parents = No | Weather Attribute

• Sunny (2 examples)

• For Parents = No | Weather = Sunny, there are 2 examples with "Tennis".

$$Gini(S) = 1 - \left[\left(\frac{2}{2} \right)^2 \right] = 0$$

Sunny

• Rainy (1 example).

• For Parents = No | Weather = Rainy, there is 1 example with "Stay In".

$$Gini(S) = 1 - \left[\left(\frac{1}{1} \right)^2 \right] = 0$$

Rainy

• Windy (2 example)

• For Parents = No | Weather = Windy, there is 1 example with "Cinema" and 1 example with "Shopping".

$$Gini(S) = 1 - \left[\left(\frac{1}{2} \right)^2 + \left(\frac{1}{2} \right)^2 \right] = 0.5$$

Windy

$$\text{Weighted Average(Parents = No | Weather)} = 0 * \left(\frac{2}{5} \right) + 0 * \left(\frac{1}{5} \right) + 0.5 * \left(\frac{2}{5} \right) = 0.2$$

G. Index

* My find for money:

Weekend	Weather	Parents	Money	Decision
W2	Sunny	No	Rich	Tennis
W5	Rainy	No	Rich	Stay In
W7	Windy	No	Poor	Cinema
W8	Windy	No	Rich	Shopping
W10	Sunny	No	Rich	Tennis

Computation of Gini Index for Parents = No | Money Attribute

• Rich (4 examples)

• For Parents = No | Money = Rich, there is 1 example with "stay in" and "Shopping" each and 2 examples of "Tennis".

$$Gini(S) = 1 - \left[\left(\frac{1}{4} \right)^2 + \left(\frac{1}{4} \right)^2 + \left(\frac{2}{4} \right)^2 \right] = 0.625$$

Computation of Gini Index for Parents = No | Money Attribute

• Poor (1 example)

• For Parents = No | Money = Poor, there is 1 example with "Cinema".

$$Gini(S) = 1 - \left[\left(\frac{1}{1} \right)^2 \right] = 0$$

$$\text{Weighted Average(Parents = No | Money)} = 0.625 * \left(\frac{4}{5} \right) + 0 * \left(\frac{1}{5} \right) = 0.5$$

(for IG $\Rightarrow$ break the largest node)

* G.I for weather $\Rightarrow$ 0.2
money $\Rightarrow$ 0.5

Smallest $\Rightarrow$ weather
 (add to tree)

Now, for Parent=No & Weather=Sunny, we have all instances as Tennis.

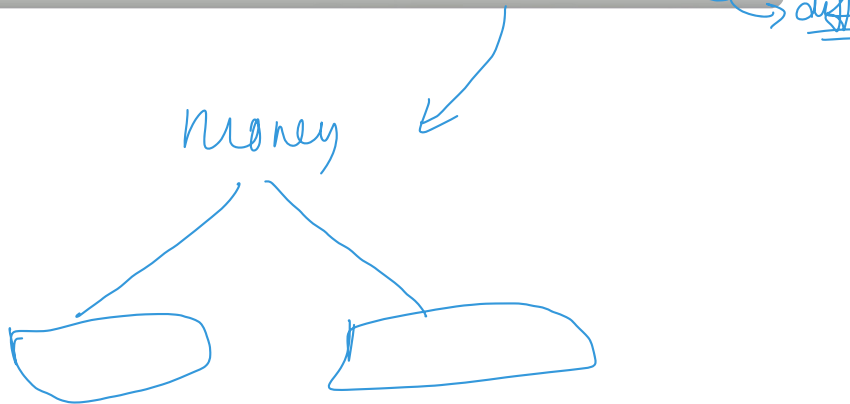
Weekend	Weather	Parents	Money	Decision
W2	Sunny	No	Rich	Tennis ✓
W10	Sunny	No	Rich	Tennis ✓

Now, for Parent=No & Weather=Windy, we need to split.

Now, for Parents=No & Weather=Rainy, we have all instances as Stay In.

Weekend	Weather	Parents	Money	Decision
W5	Rainy	No	Rich	Stay In ✓
W7	Windy	No	Poor	Cinema ✓
W8	Windy	No	Rich	Shopping ✓

(3 branches)



* Splitting continuous AHB:
 ↳ many nodes (>3)

Splitting Continuous Attribute – Decision Tree

Instance	a_1	a_2	a_3	Target Class
1	T	T	1.0	+
2	T	T	6.0	+
3	T	F	5.0	-
4	F	F	4.0	+
5	F	T	7.0	-
6	F	T	3.0	-
7	F	F	8.0	-
8	T	F	7.0	+
9	F	T	5.0	-

a_3	Target Class
1.0	+
3.0	-
4.0	+
5.0	-
5.0	-
6.0	+
7.0	-
7.0	+
8.0	-

Splitting Continuous Attribute – Decision Tree

a_3	Target Class	Split Point	Entropy	Gain
1.0	+	2.0	0.8484	0.1427
3.0	-	3.5		
4.0	+	4.5		
5.0	-			
5.0	-	5.5		
6.0	+	6.5		
7.0	-			
7.0	+	7.5		
8.0	-			

- $E = -\frac{4}{9} \log_2 \frac{4}{9} - \frac{5}{9} \log_2 \frac{5}{9} = 0.9911$ (i)
- Split Point = 2.0
- $E(a_3) = \frac{1}{9} \left[-\frac{1}{1} \log_2 \frac{1}{1} - \frac{0}{1} \log_2 \frac{0}{1} \right] + \frac{8}{9} \left[-\frac{3}{8} \log_2 \frac{3}{8} - \frac{5}{8} \log_2 \frac{5}{8} \right] = 0.8484$ (ii)
- Gain(a_3) = 0.9911 - 0.8484 = 0.1427 (iii)

Splitting Continuous Attribute – Decision Tree

a_3	Target Class	Split Point	Entropy	Gain
1.0	+	2.0	0.8484	0.1427
3.0	-	3.5		
4.0	+	4.5		
5.0	-			
5.0	-	5.5		
6.0	+	6.5		
7.0	-			
7.0	+	7.5		
8.0	-			

- $E = -\frac{4}{9} \log_2 \frac{4}{9} - \frac{5}{9} \log_2 \frac{5}{9} = 0.9911$
- Split Point = 3.5
- $E(a_3) = \frac{2}{9} \left[-\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2} \right] + \frac{7}{9} \left[-\frac{3}{8} \log_2 \frac{3}{8} - \frac{4}{8} \log_2 \frac{4}{8} \right] = 0.9885$
- Gain(a_3) = 0.9911 - 0.9885 = 0.0026

Splitting Continuous Attribute – Decision Tree

a_3	Target Class	Split Point	Entropy	Gain
1.0	+	2.0	0.8484	0.1427
3.0	-	3.5	0.9885	0.0026
4.0	+	4.5	0.9183	0.0728
5.0	-			
5.0	-	5.5	0.9839	0.0072
6.0	+	6.5	0.9728	0.0183
7.0	-			
7.0	+	7.5	0.8889	0.1022
8.0	-			

The best split for a_3 occurs
 at split point equals to 2.0

* G.I for continuous:

Splitting Continuous Attribute Gini Index Decision Tree

Annual Income	Label	Split Point	Yes	No	Gini
60	No				
70	No				
75	No	<80	0	3	
		>=80	3	4	
85	Yes				
90	Yes				
95	Yes	<97.5			
		>=97.5			
100	No				
120	No				
125	No				
220	No				

- Split Point = 80
- $$Gini(< 80) = 1 - \sum_{i=1}^c (p_i)^2$$

$$= 1 - \left(\frac{0}{3}\right)^2 - \left(\frac{3}{3}\right)^2 = 0.0$$
- $$Gini(\geq 80) = 1 - \sum_{i=1}^c (p_i)^2$$

$$= 1 - \left(\frac{3}{7}\right)^2 - \left(\frac{4}{7}\right)^2 = 0.4897$$
- $$Gini(80) = w_1 * Gini(< 80) + w_2 * Gini(\geq 80)$$

$$= \frac{3}{10} * 0.0 + \frac{7}{10} * 0.4897 = 0.3427$$

Splitting Continuous Attribute Gini Index Decision Tree

Annual Income	Label	Split Point	Yes	No	Gini
60	No				
70	No				
75	No	<80	0	3	0.3427
		>=80	3	4	
85	Yes				
90	Yes				
95	Yes	<97.5	3	3	0.30
		>=97.5	0	4	
100	No				
120	No				
125	No				
220	No				

- Split Point = 97.5
- $$Gini(< 97.5) = 1 - \sum_{i=1}^c (p_i)^2$$

$$= 1 - \left(\frac{3}{6}\right)^2 - \left(\frac{3}{6}\right)^2 = 0.5$$
- $$Gini(\geq 97.5) = 1 - \sum_{i=1}^c (p_i)^2$$

$$= 1 - \left(\frac{0}{4}\right)^2 - \left(\frac{4}{4}\right)^2 = 0.0$$
- $$Gini(97.5) = w_1 * Gini(< 97.5) + w_2 * Gini(\geq 97.5)$$

$$= \frac{6}{10} * 0.5 + \frac{4}{10} * 0.0 = 0.30$$

smaller, better